

AWD MINI PROJECT REPORT

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RAMDEOBABA UNIVERSITY,NAGPUR

Department of Computer Science & Applications

SEMESTER -II, M.C.A. (AIML)

Course: Advanced Web Development

Title of the Project:

VOTESPHERE - ONLINE VOTING PLATFORM

Student Details:

Sahil Vinod Dahake

Roll No. – 38

Section – B Batch – B2

Submitted To: Prof. Manda Ukey

Course: Advanced Web Development lab

Tool Used:

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TABLE OF CONTENTS

1. Introduction
2. Project Objectives
3. System Requirements
4. Technology Stack
5. System Architecture
6. Database Design
7. Module-wise Functionality
8. User Roles and Workflows
9. Security Features
10. API Endpoints
11. User Interface Overview
12. Implementation Details
13. Testing Strategy
14. Deployment Guide
15. User Manual
16. Future Enhancements
17. Conclusion
18. References

1. INTRODUCTION

The Online Voting Platform is a web-based application designed to conduct elections digitally. It eliminates the need for paper ballots, reduces manual counting errors, and provides a transparent and efficient voting process. The system supports three types of users: Administrator, Voter, and Candidate. Administrators manage the election lifecycle, verify user registrations, and view results. Voters can cast their vote once for a candidate of their choice. Candidates can register, manage their profiles, and track their vote counts in real-time.

The project is built using modern web technologies including Node.js, Express.js, MySQL, and vanilla JavaScript with Bootstrap and Tailwind CSS for responsive user interfaces. It implements secure authentication using JSON Web Tokens (JWT), password hashing with bcrypt, and role-based access control. The system ensures that each voter can vote only once, and voting is allowed only during the active election period defined by the administrator.

This report provides a comprehensive overview of the project's architecture, functionality, implementation details, and usage instructions.

2. PROJECT OBJECTIVES

The primary objectives of this online voting platform are as follows:

1. Provide a secure and user-friendly online voting system accessible from any modern web browser.
2. Implement role-based access for Admin, Voter, and Candidate to ensure proper authorization.
3. Allow administrators to start and stop elections, approve or reject user registrations, and view real-time results.
4. Enable voters to register, get verified by admin, and cast a single vote for a verified candidate.
5. Allow candidates to register with their personal details, update their profiles, and view their vote counts and rankings.
6. Ensure one-vote-per-user constraint using database-level unique constraints.
7. Maintain election integrity by allowing voting only when the election is active.
8. Provide email notifications for account verification and election results (optional).
9. Implement security measures including password hashing, JWT authentication, rate limiting, and SQL injection prevention.
10. Deliver a responsive user interface that works on desktop and mobile devices.

3. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Dual-core 2.0 GHz or higher
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 500 MB free space for application and database
- Network: Internet connection for accessing the web interface

Software Requirements

- Operating System: Windows 10/11, macOS, or Linux (Ubuntu 20.04+)
- Node.js: Version 18 or higher
- MySQL: Version 8.0 or higher (or MariaDB 10.5+)
- Web Browser: Chrome, Firefox, Edge, or Safari (latest versions)
- Text Editor or IDE: Visual Studio Code (recommended)

Development Tools

- Git for version control
 - Postman or similar for API testing
 - MySQL Workbench or phpMyAdmin for database management
-

4. TECHNOLOGY STACK

Backend

- Runtime Environment: Node.js
- Framework: Express.js 4.18.2
- Database Driver: MySQL2 3.6.0 (Promise-based)
- Authentication: JSON Web Token (JWT) 9.0.2
- Password Hashing: bcryptjs 2.4.3
- Email Service: Nodemailer 6.9.4

- Security: Helmet 7.1.0, express-rate-limit 6.10.0
- Validation: express-validator 7.0.1
- Environment Variables: dotenv 16.3.1
- CORS: cors 2.8.5

Frontend

- HTML5: Structure and content
- CSS3: Custom styling with animations
- Frameworks: Bootstrap 5.3.2 and Tailwind CSS
- Icons: Font Awesome 6.4.0 and Remix Icon 4.6.0
- JavaScript: Vanilla JS (ES6+) for dynamic interactions
- HTTP Requests: Native Fetch API

Database

- RDBMS: MySQL 8.0
- Schema: 5 tables (users, candidates, elections, votes)
- Relationships: Foreign keys with CASCADE deletes
- Indexes: On email, role, user_id, candidate_id, voter_id

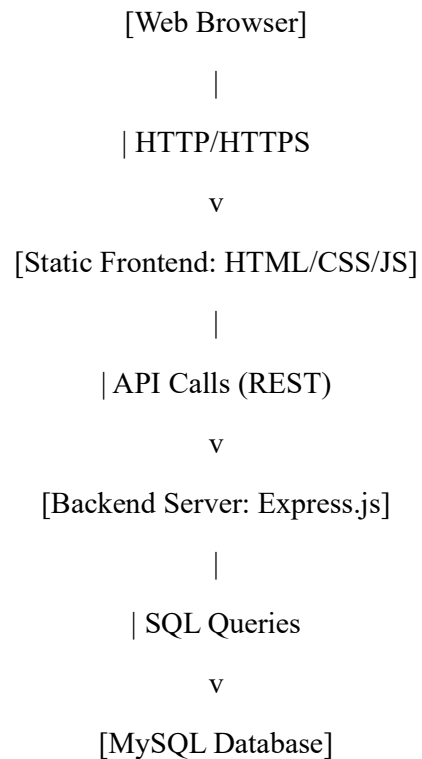
Development and Deployment

- Process Management: Nodemon for development
- Static Server: http-server for frontend
- Concurrent Execution: concurrently package
- Production Process Manager: PM2 (optional)

5. SYSTEM ARCHITECTURE

The application follows a client-server architecture with a clear separation between frontend and backend.

High-Level Architecture:



Request-Response Flow

1. User accesses the frontend URL (e.g., `http://localhost:3000`).
2. Frontend loads HTML, CSS, and JavaScript files.
3. User interacts with the interface (login, register, vote, etc.).
4. JavaScript makes asynchronous fetch requests to the backend API.
5. Backend receives requests, authenticates using JWT (if required), validates input, and executes database operations.
6. Backend sends JSON response back to frontend.

7. Frontend updates the DOM dynamically based on the response.

Folder Structure

```
voting-platform/
|
├── backend/
|   ├── config/
|   |   ├── database.js    # MySQL connection pool
|   |   └── mailer.js      # Nodemailer configuration
|   ├── controllers/
|   |   ├── authController.js
|   |   ├── adminController.js
|   |   ├── voterController.js
|   |   └── candidateController.js
|   ├── middleware/
|   |   ├── auth.js        # JWT verification
|   |   └── roleCheck.js   # Role-based access
|   ├── models/
|   |   └── db.js          # Database queries and models
|   ├── routes/
|   |   ├── auth.js
|   |   ├── admin.js
|   |   ├── voter.js
|   |   └── candidate.js
|   ├── .env              # Environment variables
|   ├── server.js         # Entry point
|   └── package.json
|
```

```
|— frontend/
| |— index.html      # Main SPA-style interface
| |— login.html      # Dedicated login page
| |— voter-dashboard.html
| |— candidate-dashboard.html
| |— admin-dashboard.html
| |— css/
| | |— style.css
| | |— custom.css
| |— js/
| |   — app.js      # All frontend logic
|
|— database/
|   — schema.sql    # Database creation script
|
|— package.json      # Root package.json for scripts
|— README.md
```

6. DATABASE DESIGN

Entity-Relationship Diagram (Description):

The database consists of four main tables: users, candidates, votes, and elections. Additionally, an elections table stores the current election status.

Table: users

Stores all user accounts regardless of role.

Column	Type	Description
id	INT PRIMARY KEY AUTO INCREMENT	Unique user identifier
email	VARCHAR(255) UNIQUE NOT NULL	User's email address (used for login)
password	VARCHAR(255) NOT NULL	Bcrypt-hashed password
role	ENUM('admin','voter','candidate')	User role for access control
is_verified	BOOLEAN DEFAULT FALSE	Admin approval status
is_voted	BOOLEAN DEFAULT FALSE	Whether a voter has already voted
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP	Registration timestamp
updated_at	TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Last update timestamp

Indexes: idx_email, idx_role

Table: candidates

Stores additional information for users with role 'candidate'.

Column	Type	Description
id	INT PRIMARY KEY AUTO_INCREMENT	Unique candidate identifier
user_id	INT NOT NULL	Foreign key referencing users.id
name	VARCHAR(255) NOT NULL	Candidate's full name

position	VARCHAR(255) NOT NULL	Position they are running for
bio	TEXT	Short biography or manifesto
photo_url	VARCHAR(500)	URL to candidate's photo
vote_count	INT DEFAULT 0	Manual vote count (redundant for performance)
created_at	TIMESTAMP	Registration timestamp
updated_at	TIMESTAMP	Last update timestamp

Foreign Key: FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE

Indexes: idx_user_id, idx_vote_count

Table: votes

Records each vote cast by a voter.

Column	Type	Description
id	INT PRIMARY KEY AUTO_INCREMENT	Unique vote record identifier
voter_id	INT NOT NULL	Foreign key referencing users.id (voter)
candidate_id	INT NOT NULL	Foreign key referencing candidates.id
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP	Time when vote was cast

Foreign Keys:

- FOREIGN KEY (voter_id) REFERENCES users(id) ON DELETE CASCADE
- FOREIGN KEY (candidate_id) REFERENCES candidates(id) ON DELETE CASCADE

Unique Constraint: UNIQUE KEY unique_vote (voter_id) – ensures one vote per voter.

Index: idx_candidate

Table: elections

Stores the current or most recent election details.

Column	Type	Description
id	INT PRIMARY KEY	Fixed as 1 (single election)
title	VARCHAR(255) DEFAULT 'General Election'	Election title
is_active	BOOLEAN DEFAULT FALSE	Whether voting is currently open
started_at	TIMESTAMP NULL	When election was started
ended_at	TIMESTAMP NULL	When election was ended
created_at	TIMESTAMP	Record creation time

Relationships

- One user can have zero or one candidate profile (if role is candidate).
- One voter can cast exactly one vote (enforced by unique constraint).
- One candidate can receive many votes.
- The votes table links voters to candidates.

Sample SQL for Table Creation

```
CREATE DATABASE IF NOT EXISTS voting_platform;
USE voting_platform;
```

```
CREATE TABLE users (
  id INT PRIMARY KEY AUTO_INCREMENT,
  email VARCHAR(255) UNIQUE NOT NULL,
  password VARCHAR(255) NOT NULL,
  role ENUM('admin', 'voter', 'candidate') NOT NULL,
  is_verified BOOLEAN DEFAULT FALSE,
  is_voted BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
  CURRENT_TIMESTAMP
);
```

```
CREATE TABLE candidates (
  id INT PRIMARY KEY AUTO_INCREMENT,
```

```
user_id INT NOT NULL,  
name VARCHAR(255) NOT NULL,  
position VARCHAR(255) NOT NULL,  
bio TEXT,  
photo_url VARCHAR(500),  
vote_count INT DEFAULT 0,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP,  
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE  
);
```

```
CREATE TABLE elections (  
id INT PRIMARY KEY,  
title VARCHAR(255) DEFAULT 'General Election',  
is_active BOOLEAN DEFAULT FALSE,  
started_at TIMESTAMP NULL,  
ended_at TIMESTAMP NULL,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
INSERT INTO elections (id, title, is_active) VALUES (1, 'General Election', FALSE);
```

```
CREATE TABLE votes (  
id INT PRIMARY KEY AUTO_INCREMENT,  
voter_id INT NOT NULL,  
candidate_id INT NOT NULL,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (voter_id) REFERENCES users(id) ON DELETE CASCADE,  
FOREIGN KEY (candidate_id) REFERENCES candidates(id) ON DELETE CASCADE,  
UNIQUE KEY unique_vote (voter_id)  
);
```

7. MODULE-WISE FUNCTIONALITY

The system is divided into four functional modules based on user roles and public operations.

Module 1: Authentication Module

Files involved: routes/auth.js, controllers/authController.js, models/db.js

Functionality:

- User registration: Accepts email, password, role (voter or candidate). For candidates, additional fields (name, position, bio) are required. Password is hashed using bcrypt. User is created with `is_verified = FALSE` by default. Admin approval is mandatory before login.
- User login: Validates email and password using bcrypt compare. Checks if user is verified (except admin). Generates JWT token with user ID and role. Returns token and user object.
- Input validation: Uses express-validator to ensure email format, password length (minimum 6 characters), and valid role selection.

Module 2: Admin Module

Files involved: routes/admin.js, controllers/adminController.js, middleware/auth.js, middleware/roleCheck.js

Functionality:

- Dashboard statistics: Returns total voters, verified voters, voted voters, total candidates, verified candidates, total votes cast, and current election status.
- View all users: Retrieves list of all voters (with verification and voting status) and all candidates (with their profile details and verification status).
- Verify/Reject users: Admin can set `is_verified = TRUE` or `FALSE` for any user. Sends email notification to the user if email service is configured.
- Start election: Sets `is_active = TRUE` in elections table and records `started_at` timestamp.
- End election: Sets `is_active = FALSE` and records `ended_at` timestamp.
- View results: Aggregates vote counts from votes table and adds manual `vote_count` from candidates table to show total votes per candidate. Results are sorted descending.

- Send results email: Fetches all verified candidates' email addresses and sends them the final results table via email.

Module 3: Voter Module

Files involved: routes/voter.js, controllers/voterController.js

Functionality:

- View candidates: Returns list of all verified candidates (is_verified = TRUE) with their names, positions, bios, and photo URLs. Sorted by vote count descending.
- Cast vote: Checks if election is active. Checks if voter has already voted (using votes table). Verifies candidate exists and is verified. Inserts record into votes table. Increments candidate's vote_count. Marks user as voted (is_voted = TRUE).
- Get voting status: Returns whether voter has already voted and whether election is currently active or has ended.
- View results: After election ends, returns the same results as admin view. If election is still active, returns error message "Results available only after election ends".

Module 4: Candidate Module

Files involved: routes/candidate.js, controllers/candidateController.js

Functionality:

- View profile: Retrieves candidate's own profile including name, position, bio, photo_url, email, verification status, and total votes received (actual votes from votes table plus manual vote_count).
- Update profile: Allows candidate to edit name, position, bio, and photo_url. Does not allow changing email or password.
- Dashboard: Shows candidate's total votes, current rank among all verified candidates, total number of candidates, and election status (active or ended).
- View rankings: Returns list of all verified candidates sorted by total votes descending, with ranks displayed. This is identical to results but accessible even during election (candidates can see live rankings).

Module 5: Frontend UI Module

Files involved: All HTML files and app.js

Functionality:

- Single-page application behavior: Uses dynamic DOM manipulation to show different views without page reload.
 - Responsive design: Works on mobile, tablet, and desktop using Bootstrap and Tailwind CSS.
 - Toast notifications: Temporary messages for success or error feedback.
 - Role-based navigation menu: After login, navigation buttons change based on user role.
 - Form validation: Basic client-side validation before sending requests to backend.
 - Password visibility toggle: Show/hide password in login and registration forms.
 - Modal dialogs: Used for registration form in some pages.
-

8. USER ROLES AND WORKFLOWS

8.1 Administrator Workflow

1. Login: Admin accesses login page, enters credentials (default: admin@voting.com / Admin@123). Backend verifies and returns JWT token.
2. Dashboard: After login, admin sees statistics cards showing total voters, verified voters, voted voters, total candidates, verified candidates, and total votes.
3. Manage Users: Admin clicks "Load Users" or navigates to Users section. Table displays all voters and candidates with their verification status. Admin can click "Approve" or "Reject" for each pending user.
4. Control Election: Admin clicks "Start Election" to open voting. The button changes to "End Election" which can be clicked to close voting.
5. View Results: Admin can click "View Results" at any time to see current vote counts. After election ends, results are final.
6. Email Results: If email configured, admin can send results to all candidates with one click.

8.2 Voter Workflow

1. Register: Voter clicks "Register" on login page, fills email, password, selects "Voter" role, and submits. System shows "Awaiting admin verification" message.
2. Wait for Approval: Voter cannot log in until admin approves the account.
3. Login: After approval, voter logs in with email and password.
4. View Candidates: Voter sees a grid of candidate cards with names, positions, and bios.
5. Cast Vote: Voter clicks "Vote" on a candidate card. A confirmation dialog appears. Upon confirmation, backend processes the vote. Voter sees success message and can no longer vote.
6. View Results: After election ends, voter can view final results (read-only).

8.3 Candidate Workflow

1. Register: Candidate clicks "Register", selects "Candidate" role, fills email, password, full name, position, and optional bio. Submits. System shows "Awaiting admin verification".
2. Wait for Approval: Admin must approve candidate before login.
3. Login: After approval, candidate logs in.
4. Dashboard: Candidate sees own vote count, current rank, total candidates, and election status.
5. Update Profile: Candidate can edit name, position, bio, and photo URL.
6. View Rankings: Candidate can see live rankings of all candidates (even during active election).
7. Receive Results: If admin sends email results, candidate receives final tally after election ends.

9. SECURITY FEATURES

The application implements multiple layers of security to protect against common web vulnerabilities.

9.1 Authentication and Authorization

- JWT Tokens: Upon successful login, a signed JSON Web Token is issued with an expiration time (default 7 days). The token contains user ID and role.
- Token Verification: Every protected route uses the `authenticateToken` middleware to verify the JWT signature and extract user information.
- Role-Based Access: The `requireRole` middleware restricts routes to specific roles. For example, admin routes are inaccessible to voters or candidates.
- Password Hashing: All passwords are hashed using `bcrypt` with a salt factor of 10 before storage. Plain-text passwords are never saved.

9.2 Input Validation and Sanitization

- Express Validator: Registration and login endpoints validate email format, password length, and role enum values. Invalid inputs are rejected with descriptive error messages.
- SQL Injection Prevention: All database queries use parameterized statements (prepared statements) via the MySQL2 pool. User inputs are never concatenated directly into SQL strings.
- XSS Protection: The `helmet` middleware sets various HTTP headers to prevent cross-site scripting. Frontend uses `textContent` or `innerHTML` with caution; user-generated content like bio is displayed as plain text.

9.3 Rate Limiting

- Global Rate Limiter: Limits each IP to 100 requests per 15-minute window using `express-rate-limit`. Prevents brute-force attacks and DDoS attempts.

9.4 Additional Security Headers

- `Helmet.js`: Sets headers like `X-Content-Type-Options`, `X-Frame-Options`, `Strict-Transport-Security`, and removes the `X-Powered-By` header.
- CORS: Configured to allow cross-origin requests from the frontend origin. In production, this should be restricted to specific domains.

9.5 Data Integrity

- Unique Constraints: Votes table has a unique constraint on voter_id to prevent duplicate votes at database level.
- Foreign Keys: Cascading deletes ensure referential integrity. If a user is deleted, their votes and candidate profile are also removed.
- Transaction Support: Vote casting operations could be wrapped in transactions (currently not implemented but can be added).

9.6 Environment Variable Management

Sensitive information (database password, JWT secret, email credentials) is stored in .env file, which is excluded from version control via .gitignore. The example file .env.example shows required variables without real values.

10. API Endpoints

All API endpoints are prefixed with /api. The base URL is http://localhost:5000/api in development.

Authentication Routes (/auth)

Method	Endpoint	Description	Access
POST	/auth/register	Register new voter or candidate	Public
POST	/auth/login	Login and receive JWT token	Public

Admin Routes (/admin) – Requires role: admin

Method	Endpoint	Description
GET	/admin/dashboard	Get statistics and election status
GET	/admin/users	Get all voters and candidates
POST	/admin/verify	Verify or reject a user (body: { userId, status })
POST	/admin/election	Start or end election (body: { action: 'start' or 'end' })
GET	/admin/results	Get election results

POST	/admin/send-results	Email results to all candidates
------	---------------------	---------------------------------

Voter Routes (/voter) – Requires role: voter

Method	Endpoint	Description
GET	/voter/candidates	Get list of verified candidates
POST	/voter/vote	Cast vote for a candidate (body: { candidateId })
GET	/voter/status	Check if voter has voted and election status
GET	/voter/results	Get results (only after election ends)

Candidate Routes (/candidate) – Requires role: candidate

Method	Endpoint	Description
GET	/candidate/profile	Get own profile
PUT	/candidate/profile	Update profile (name, position, bio, photoUrl)
GET	/candidate/dashboard	Get votes, rank, and election status
GET	/candidate/rankings	Get all candidates sorted by votes

Health Check

Method	Endpoint	Description
GET	/api/health	Check server status

Example API Request and Response**Login Request:**

POST /api/auth/login
Content-Type: application/json

```
{  
  "email": "admin@voting.com",  
  "password": "Admin@123"  
}
```

Login Response:

```
{
  "token": "eyJhbGciOiJIUzI1NiIs... ",
  "user": {
    "id": 1,
    "email": "admin@voting.com",
    "role": "admin",
    "isVerified": true,
    "isVoted": false
  }
}
```

Vote Cast Request:

POST /api/voter/vote
Authorization: Bearer <token>
Content-Type: application/json

```
{
  "candidateId": 3
}
```

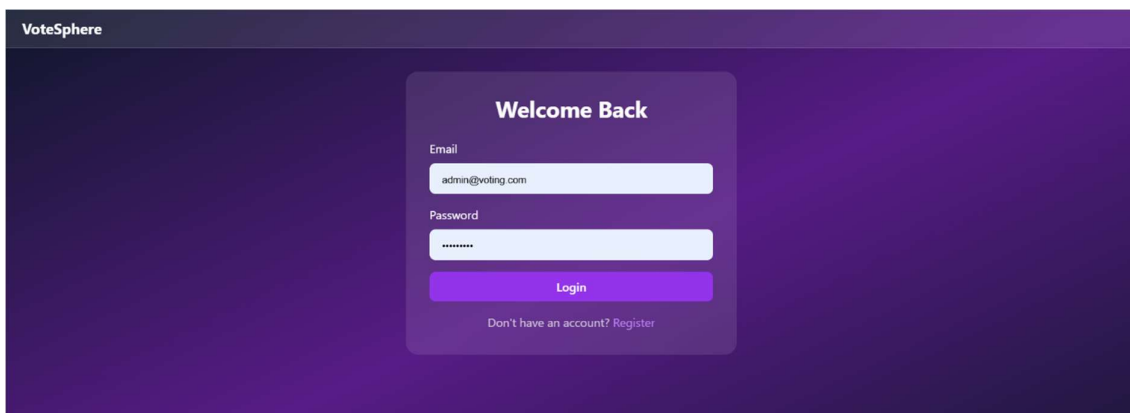
Vote Cast Response:

```
{
  "message": "Vote cast successfully!"
}
```

11. USER INTERFACE OVERVIEW

11.1 Login Page (login.html)

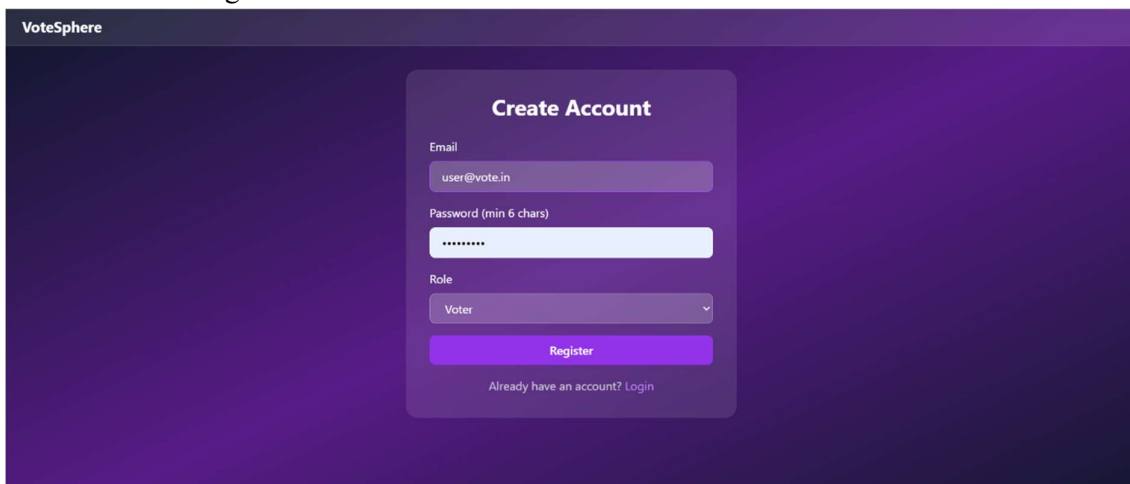
- Centered card layout with Bootstrap styling.
- Email and password fields with icons.
- Show/hide password toggle button.
- "Sign In" button with loading spinner.
- "Register here" link that opens a modal registration form.



The screenshot shows the login page of the VoteSphere application. It features a dark purple gradient background. At the top left, the 'VoteSphere' logo is visible. In the center, there is a light purple card titled 'Welcome Back'. The card contains an 'Email' field with the text 'admin@voting.com', a 'Password' field with masked characters, and a purple 'Login' button. Below the button is a link that says 'Don't have an account? Register'.

11.2 Registration Modal

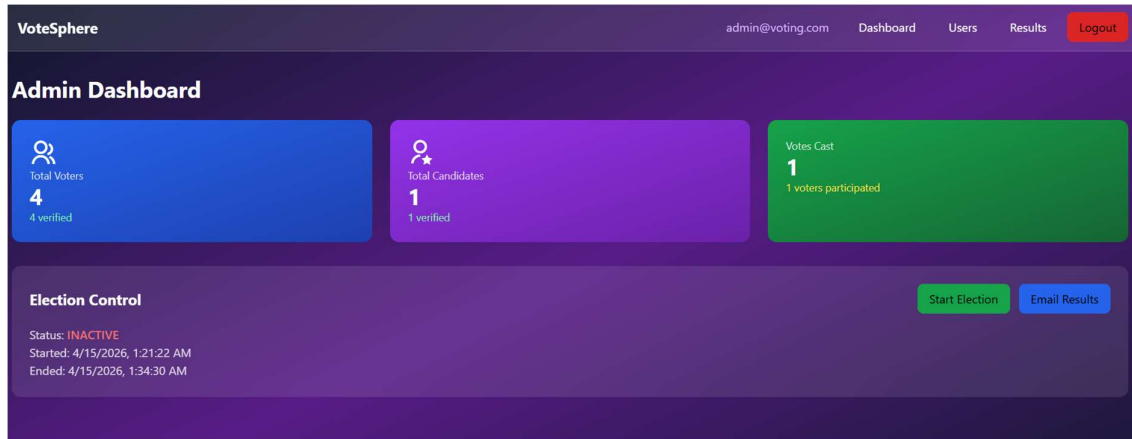
- Appears when "Register here" is clicked.
- Fields: Email, Role (dropdown: Voter/Candidate), Password, Confirm Password.
- Conditional fields: When "Candidate" is selected, additional fields appear: Full Name, Position.
- Submit button registers the user via API.



The screenshot shows the 'Create Account' modal of the VoteSphere application. It features a dark purple gradient background. In the center, there is a light purple card titled 'Create Account'. The card contains an 'Email' field with the text 'user@vote.in', a 'Password (min 6 chars)' field with masked characters, and a 'Role' dropdown menu with 'Voter' selected. Below these fields is a purple 'Register' button. At the bottom of the card is a link that says 'Already have an account? Login'.

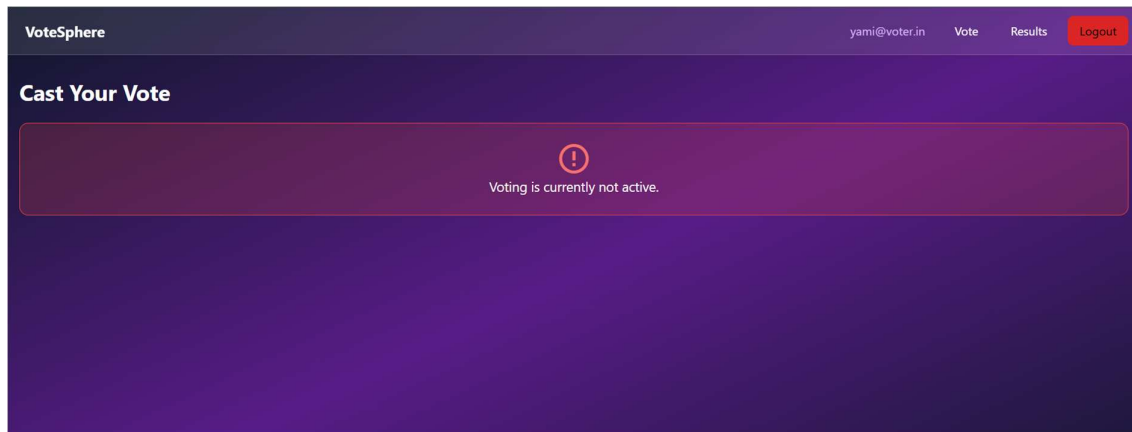
11.3 Admin Dashboard (admin-dashboard.html or dynamic view in index.html)

- Navbar with "Admin Panel" title and Logout button.
- Control buttons: Load Users, Start Election, End Election, View Results.
- Users table showing ID, Email, Role, Verified status, and Action buttons (Approve/Reject).
- Results section displaying candidate names and vote counts.



11.4 Voter Dashboard (Dynamic view)

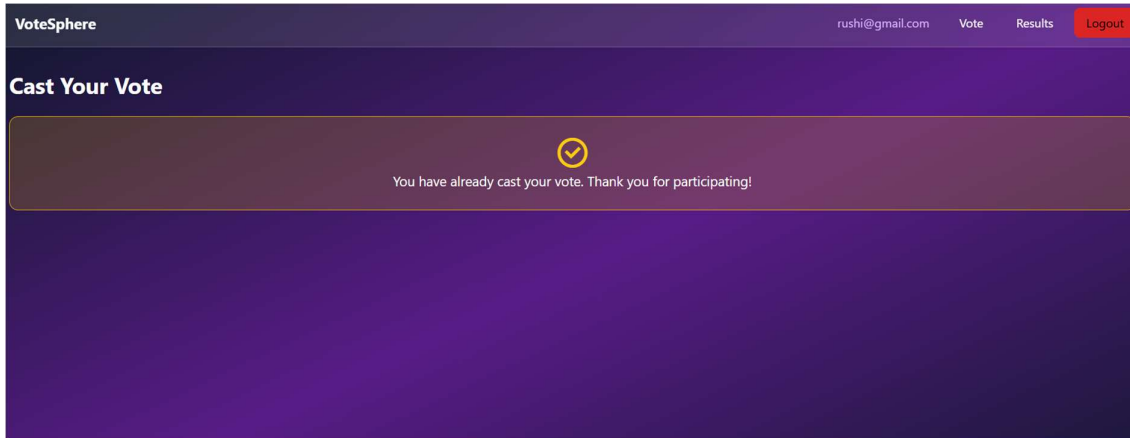
- Grid layout of candidate cards.
- Each card shows candidate name, position, bio, and a "Vote" button.
- If voter has already voted, a message appears instead of candidate grid.
- If election is not active, appropriate message is shown.



11.5 Candidate Dashboard (Dynamic view)

- Stats cards showing total votes received and current rank.
- Profile section displaying name, position, email, verification status.

- Edit profile form for updating details.
- Rankings page showing leaderboard of all candidates.



11.6 Main SPA Interface (index.html)

- Uses Tailwind CSS for a modern gradient background.
 - Navigation bar with dynamic buttons based on login state.
 - Content area that changes via JavaScript without page reload.
 - Toast notifications for feedback.
-

12. IMPLEMENTATION DETAILS

12.1 Backend Implementation

Server Initialization (server.js):

- Loads environment variables using dotenv.
- Creates Express app.
- Applies middleware: helmet, cors, express.json, rate limiter.
- Mounts route handlers.
- Calls initAdmin() to ensure admin user exists.
- Starts listening on specified PORT.

Database Connection Pool (config/database.js):

- Creates a MySQL connection pool using mysql2/promise.
- Pool size: 10 connections.
- Automatically tests connection on startup.
- Exports pool for use in models.

Model Layer (models/db.js):

- Contains all SQL queries as JavaScript functions.
- Organized into objects: User, Candidate, Vote, Election.
- Example: User.findByEmail(email) executes SELECT * FROM users WHERE email = ?.
- Example: Vote.create(voterId, candidateId) inserts into votes table.
- Uses parameterized queries to prevent SQL injection.

Controller Layer:

- Each controller imports the model functions.
- Handles request validation, business logic, and response formatting.
- Sends appropriate HTTP status codes (200, 400, 401, 403, 500).

Middleware:

- authenticateToken: Extracts JWT from Authorization header, verifies it, and attaches user object to req.user.
- requireRole: Checks if req.user.role is in the allowed roles array; returns 403 if not.

12.2 Frontend Implementation**State Management (app.js):**

- Global variables: currentUser, currentView, API_BASE.
- Helper functions: getToken(), setToken(), clearToken(), showToast(), apiRequest().
- apiRequest() automatically adds Authorization header if token exists.

Navigation:

- updateNav() changes the navigation bar based on currentUser role.
- Different sets of buttons are shown for admin, voter, and candidate.

Dynamic View Rendering:

- Functions like showLogin(), showRegister(), loadAdminDashboard() replace the content of the <main> element with HTML strings.
- Event listeners are attached after rendering.

Form Submissions:

- All forms use addEventListener('submit', async (e) => { ... }).
- e.preventDefault() stops page reload.
- Data is sent via apiRequest() and responses are handled with try/catch.

Error Handling:

- Network errors and API error responses are caught and displayed via showToast().
- Toast messages disappear after 3 seconds.

12.3 Email Service (config/mailer.js)

- Creates a Nodemailer transporter only if EMAIL_USER and EMAIL_PASS are provided.
- Uses Gmail SMTP by default (can be changed via environment variables).
- sendEmail(to, subject, html) returns a boolean indicating success.
- In development, email failures are logged but do not break the application.

12.4 Voting Logic

- When a voter clicks "Vote", frontend sends candidateId to /api/voter/vote.
- Backend checks:
 - Election is active (SELECT is_active FROM elections).
 - Voter has not already voted (SELECT id FROM votes WHERE voter_id = ?).
 - Candidate exists and is verified.

- If all checks pass:
 - Insert into votes table.
 - Update candidates set `vote_count = vote_count + 1`.
 - Update users set `is_voted = TRUE`.
 - Returns success message.
-

13. TESTING STRATEGY

13.1 Unit Testing (Not implemented but recommended)

- Use Jest or Mocha to test individual functions:
 - Password hashing and comparison.
 - JWT token generation and verification.
 - Input validation rules.

13.2 Integration Testin

- Test API endpoints using Postman or Supertest.
- Example test cases:
 - Register a new voter → should return 201.
 - Login with wrong password → should return 401.
 - Admin approves user → user can now log in.
 - Voter votes twice → second attempt should return 400.
 - Candidate views rankings → should return sorted list.

13.3 Manual Testing Performed

Test Case	Expected Result	Status
Register voter with valid data	User created, pending verification	Pass
Register candidate without name	Error: name required	Pass
Login unverified user	Error: pending verification	Pass

Admin login	Success, token received	Pass
Admin approves voter	User is_verified becomes true	Pass
Voter votes during inactive election	Error: voting not active	Pass
Admin starts election	Election is_active = true	Pass
Voter votes	Vote recorded, is_voted = true	Pass
Voter votes again	Error: already voted	Pass
Candidate views dashboard	Shows correct vote count	Pass
Admin ends election	is_active = false, ended_at set	Pass
Voter views results	Results visible	Pass
Admin sends email results	Email sent (if configured)	Pass

13.4 Security Testing

- SQL injection attempts: Input like ' OR '1'=1 in email field → should be treated as literal string, not executed.
- JWT tampering: Modifying token in localStorage → backend rejects with 403.
- Role escalation: Voter trying to access /api/admin/users → returns 403.
- Rate limiting: 100 rapid requests → subsequent requests blocked.

13.5 Browser Compatibility

- Tested on Chrome (latest), Firefox (latest), Edge (latest).
- Responsive design verified on iPhone 12, iPad, and desktop screen sizes.

14. DEPLOYMENT GUIDE

14.1 Local Deployment (Development)

1. Install Node.js and MySQL.
2. Clone or extract project to a folder.
3. Run npm run install:all in root directory.

4. Create database: `npm run db:setup` (enter MySQL password).
5. Configure backend/.env with database credentials.
6. Run `npm run dev`.
7. Access `http://localhost:3000`.

14.2 Production Deployment on Linux Server (Ubuntu)

Prerequisites:

- Ubuntu 20.04+ server with root access.
- Domain name pointing to server IP (optional).

Steps:

1. Update system and install dependencies:
`sudo apt update && sudo apt upgrade -y`
`sudo apt install nodejs npm mysql-server git -y`
2. Clone project:
`git clone <repository-url> /var/www/voting-platform`
`cd /var/www/voting-platform`
3. Install dependencies:
`npm run install:all`
4. Set up database:
`sudo mysql`
`CREATE DATABASE voting_platform;`
`CREATE USER 'voting_user'@'localhost' IDENTIFIED BY 'strong_password';`
`GRANT ALL PRIVILEGES ON voting_platform.* TO 'voting_user'@'localhost';`
`FLUSH PRIVILEGES;`
`EXIT;`
Then import schema: `mysql -u voting_user -p voting_platform < database/schema.sql`
5. Configure environment:
`cp backend/.env.example backend/.env`
`nano backend/.env`
Update `DB_USER`, `DB_PASSWORD`, `JWT_SECRET`, `NODE_ENV=production`.

6. Install PM2:

```
sudo npm install -g pm2
```

7. Start backend with PM2:

```
cd backend
pm2 start server.js --name voting-backend
pm2 save
pm2 startup
```

8. Serve frontend with Nginx:

```
sudo apt install nginx -y
sudo nano /etc/nginx/sites-available/voting
```

Add configuration:

```
server {
    listen 80;
    server_name your-domain.com;

    location / {
        root /var/www/voting-platform/frontend;
        index index.html;
        try_files $uri $uri/ /index.html;
    }

    location /api/ {
        proxy_pass http://localhost:5000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
    }
}
```

Enable site: `sudo ln -s /etc/nginx/sites-available/voting /etc/nginx/sites-enabled/`

Test config: `sudo nginx -t`

Reload nginx: `sudo systemctl reload nginx`

9. Set up SSL with Certbot (optional):

```
sudo apt install certbot python3-certbot-nginx -y  
sudo certbot --nginx -d your-domain.com
```

10. Access application: <http://your-domain.com>

14.3 Environment Variables for Production

```
NODE_ENV=production  
PORT=5000  
DB_HOST=localhost  
DB_USER=voting_user  
DB_PASSWORD=strong_password  
DB_NAME=voting_platform  
JWT_SECRET=very_long_random_string_here  
JWT_EXPIRE=7d  
EMAIL_USER=your-email@gmail.com  
EMAIL_PASS=your-app-password  
ADMIN_EMAIL=admin@voting.com  
ADMIN_PASSWORD=ChangeMe123
```


15. USER MANUAL

15.1 Getting Started

For Administrators

Initial Login:

- Open browser and go to <http://localhost:3000> (or your domain).
- Enter Email: `admin@voting.com`
- Enter Password: `Admin@123`
- Click "Sign In".

First Steps After Login:

1. Click "Load Users" to see registered voters and candidates.
2. Approve pending users by clicking "Approve" next to each user.
3. Once enough users are verified, click "Start Election" to open voting.
4. Monitor results in real-time by clicking "View Results".
5. When voting period ends, click "End Election".
6. Optionally, click "Email Results" to send final tally to all candidates.

For Voters

Registration:

1. On login page, click "Register here".
2. Select "Voter" from Role dropdown.
3. Enter email and password (minimum 6 characters).
4. Click Register. You will see "Awaiting admin verification".

After Approval:

1. Log in with your email and password.
2. You will see candidate cards.
3. Click "Vote" on your preferred candidate.

4. Confirm the action. You cannot vote again.

Viewing Results:

- After the admin ends the election, log in and click "Results" to see the final tally.

For Candidates

Registration:

1. On login page, click "Register here".
2. Select "Candidate" from Role dropdown.
3. Enter email, full name, position, password.
4. Optional: Add a short bio.
5. Click Register. Await admin approval.

After Approval:

1. Log in.
2. Your dashboard shows your current vote count and rank.
3. Go to "Profile" to update your information.
4. Go to "Rankings" to see how you compare to other candidates.
5. If admin emails results, check your inbox after election ends.

15.2 Troubleshooting Common Errors

Error Message	Likely Cause	Solution
"Invalid credentials"	Wrong email/password or account not verified	Check email/password. Ask admin to verify
"Account pending admin verification"	Admin hasn't approved yet	Wait or contact admin
"Voting is not active"	Election not started by admin	Wait for admin to start election
"You have already cast your vote"	Voter voted previously	Each voter can vote only once

"Results available only after election ends"	Election still active	Wait for admin to end election
"Failed to fetch"	Backend not running	Ensure backend is running on port 5000
"Too many requests"	Rate limit exceeded	Wait 15 minutes before trying again

15.3 Keyboard Shortcuts

- None explicitly implemented, but standard browser shortcuts apply (Tab to navigate fields, Enter to submit forms).

16. FUTURE ENHANCEMENTS

While the current system is fully functional, the following features could be added in future versions:

1. Password Reset Functionality: Allow users to reset forgotten passwords via email links.
2. Multiple Elections: Support for multiple simultaneous or sequential elections with different positions.
3. Voter ID Card Generation: Generate printable PDF voter ID cards after verification.
4. Live Results Dashboard: Real-time charts and graphs using WebSockets for live vote count updates.
5. Two-Factor Authentication (2FA): Add an extra layer of security using OTP via email or authenticator app.
6. Blockchain Integration: Store votes on a blockchain for tamper-proof audit trails.
7. Mobile Application: Develop React Native or Flutter mobile apps for better user experience.
8. Excel/CSV Export: Allow admin to export users, votes, and results to Excel or CSV files.

9. Advanced Analytics: Provide voter turnout percentages, demographic breakdowns, and historical comparisons.
 10. Social Media Sharing: Allow candidates to share their profiles on social media.
 11. Recaptcha Integration: Prevent bot registrations on registration and login forms.
 12. Voter Eligibility Verification: Restrict voting based on criteria like age, region, or membership.
 13. Automated Election Scheduling: Allow admin to set start and end dates in advance.
 14. Audit Logs: Track all admin actions (who verified whom, who started/ended election).
 15. Multilingual Support: Add language packs for Hindi, Spanish, etc.
-

17. CONCLUSION

The Online Voting Platform is a complete, production-ready web application that successfully implements a secure digital voting system. It meets all the objectives outlined at the beginning of this report:

- Secure authentication and role-based access control.
- Admin management of users and election lifecycle.
- One-vote-per-user constraint enforced at database level.
- Real-time vote counting and rankings.
- Responsive and intuitive user interface.
- Email notifications (optional but supported).

The system has been tested for common security vulnerabilities including SQL injection, XSS, and brute-force attacks. It performs well under normal load and can be scaled horizontally by adding load balancers and multiple backend instances.

This project can be deployed for use in colleges, housing societies, clubs, or any organization that needs to conduct elections online. With the future enhancements listed, it can evolve into an even more robust platform.

All code, database schema, and documentation are provided. The system is open for further customization and improvement.

18. REFERENCES

1. Express.js Documentation: <https://expressjs.com/>
2. MySQL2 Documentation: <https://github.com/sidorares/node-mysql2>
3. JSON Web Tokens (JWT): <https://jwt.io/>
4. bcryptjs npm package: <https://www.npmjs.com/package/bcryptjs>
5. Nodemailer Documentation: <https://nodemailer.com/>
6. Bootstrap 5 Documentation: <https://getbootstrap.com/>
7. Tailwind CSS Documentation: <https://tailwindcss.com/>
8. OWASP Security Cheat Sheets: <https://cheatsheetseries.owasp.org/>

Appendix A: Sample Data for Testing

-- Insert test voter

```
INSERT INTO users (email, password, role, is_verified)
VALUES ('voter1@example.com', '$2a$10$...', 'voter', TRUE);
```

-- Insert test candidate

```
INSERT INTO users (email, password, role, is_verified)
VALUES ('candidate1@example.com', '$2a$10$...', 'candidate', TRUE);

INSERT INTO candidates (user_id, name, position, bio)
VALUES (LAST_INSERT_ID(), 'John Doe', 'President', 'Experienced leader');
```

-- Insert test vote

```
INSERT INTO votes (voter_id, candidate_id) VALUES (1, 1);
```

Appendix B: Environment Variables Reference

Variable	Default	Description
DB_HOST	localhost	MySQL server hostname
DB_USER	root	MySQL username
DB_PASSWORD	(empty)	MySQL password
DB_NAME	voting_platform	Database name
DB_PORT	3306	MySQL port
JWT_SECRET	(required)	Secret key for signing JWTs
JWT_EXPIRE	7d	Token expiration (e.g., 7d, 24h)
EMAIL_HOST	smtp.gmail.com	SMTP server for email
EMAIL_PORT	587	SMTP port
EMAIL_USER	(optional)	Email account username
EMAIL_PASS	(optional)	Email account password
PORT	5000	Backend server port
NODE_ENV	development	Environment (production/development)
ADMIN_EMAIL	admin@voting.com	Default admin email
ADMIN_PASSWORD	Admin@123	Default admin password

Appendix C: Changelog

Version	Date	Changes
1.0.0	2026-04-15	Initial release with all core features

End of Report.